

Alex Lee

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EDUCATION

University of New South Wales

Sydney, Australia

Bachelor of Computer Science

Graduation: May 2026

- WAM: 81 — 2025 Academic Year WAM: 87
- COMP3311 - 91 HD, COMP3231 - 89 HD, COMP1511 - 88 HD, COMP3900 - 85 HD
- Exchange Program to Seoul National University (*South Korea*) in 2024 T3 (Translated WAM: 83)
- 2024 New Colombo Plan Scholarship (AUD \$7000), 2025 Taste of Research Scholarship (AUD \$6000)
- UniHack 2025 Hackathon Best Design Award & Social Impact Award (Only team out of 135 teams to be awarded 2 out of a total of 11 awards, with a total of 650+ participants)
- Computer Science Mentor for Colombo House (May 2024 - Aug 2024)
- Trainee Lead for UNSW Development Society (Mar 2025 - May 2025)

EXPERIENCE

Software Engineer Intern

Nov 2025 – Present

Google LLC

Sydney, Australia

- Google Maps SRE Intern, Geo-Data Team
- Maintaining quality of Google Maps Data across the world, with geographical data, user-generated contents, etc.
- Using and learning Java, Go, C/C++, Protobufs, gRPCs and other various tools in a real-life environment.

Software Engineer Intern

Jun 2025 – Nov 2025

TikTok Pte. Ltd.

Sydney, Australia

- Trust & Safety Engineering in Test, Engagement Safety (Direct Messages)
- Enhanced Python QA frameworks to test Go microservices and drove DM Safety stability & coverage metrics from 6th to 1st out of 14 Trust and Safety subteams, maintaining RPC test stability rates of 99%+ every week.
- Scaled testing infrastructure, developed internal tools, created testing pipelines for the stability of millions of TikTok DM users, cutting API latencies by up to 83% and test runtimes by 81%.
- Occasionally participated in on-calls for priority bugs, and created reports for weekly meetings every week.

Tutor (Teaching Assistant)

Feb 2024 – Present

UNSW Computer Science & Engineering

Sydney, Australia

- COMP2041 (Software Construction [*Linux, Shell, Python*]) Tutor: Overall rating 5.88/6.00 (avg: 5.39/6.00).
- COMP3311 (Database Systems [*SQL, Database Design*]) Tutor: Overall rating 5.67/6.00 (avg: 5.39/6.00).

PROJECTS

SmartTransit | *Python, Typescript, React Native, SQL, AWS (RDS), Algorithms, Data Structures, Stripe, Docker*

- Created a bus transportation app where bus routes dynamically change according to real-time customer needs.
- Used Typescript React Native for the frontend, and Flask for the backend, with infrastructure built on AWS.
- Integrated Stripe for secure payments, Google Maps API for live tracking, and JWT for robust user authentication.

Redis Clone | *C, C++, Socket Programming, Networks, Data Structures, Operating Systems, Redis*

- Built a high-performance Redis clone from scratch, featuring a non-blocking, event-driven architecture using polling for scalable multi-client handling, efficient I/O, and minimal syscall overhead
- Implemented custom protocols for client-server communication, with RESP-style command parsing.
- Explored various client connection models, including multithreading, event-driven polling, and WebSockets to compare scalability, latency, and system resource usage.

Distributed File Storage System | *Go, Networks, Cryptography, Concurrency, Caching, File Systems*

- Built a decentralized, content-addressable file storage system in Go, enabling P2P file sharing and retrieval.
- Implemented protocols for secure data streaming, handshake verification, and efficient large-file transfers.
- Optimised performance with content-based addressing, caching, and encryption, achieving scalable, fault-tolerant distributed storage across nodes.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Go, SQL, JavaScript, TypeScript

Systems: Database Systems, Computer Networks, Operating Systems, Distributed Systems

Tools: Flask, React Native, AWS, Docker, Redis, gRPC, Protobuf

Misc: Data Structures, Algorithms, System Design, Pytest, Git, CI/CD, FastAPI, Tailwind